

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to automate the credit card approval process using Machine Learning. Traditional approval methods are time-consuming and involve manual verification. This project predicts whether a customer's credit card application is likely to be approved based on their demographic, financial, and employment information. The system improves decision-making speed, accuracy, and consistency while reducing manual effort.

Project Overview	
Objective	To develop a Machine Learning-based web application that predicts credit card approval by analyzing applicant information such as income, age, employment status, education, family status, and occupation.
Scope	The project focuses on: • Data preprocessing • Feature engineering • Handling missing values • Balancing the dataset using SMOTE • Training multiple Machine Learning models • Comparing model performance • Deploying the best-performing model using Flask
Problem Statement	
Description	Manual credit card approval involves extensive document verification and human judgment, making the process slow and prone to inconsistency. Financial institutions require an intelligent system that can provide quick and reliable approval decisions.

Impact	Implementing this system will: • Reduce approval time • Improve prediction accuracy • Minimize human errors • Provide consistent decisions • Assist financial institutions in risk assessment
Proposed Solution	
Approach	The system preprocesses applicant data, performs feature engineering, encodes categorical variables, balances the dataset using SMOTE, and trains multiple Machine Learning models. The best-performing model (Random Forest) is deployed as a Flask web application to predict approval in real time.
Key Features	• Machine Learning-based credit card approval prediction • Automatic data preprocessing • Feature Engineering (Age and Years Employed) • Missing value handling • Label Encoding for categorical variables • SMOTE for balancing the dataset • Multiple model comparison • Random Forest best model selection • Prediction confidence score • Responsive Flask web interface

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (Dual/Quad Core)
Memory	RAM specifications	8 GB

Storage	Disk space for dataset, trained model, and project files	10 GB Free SSD/HDD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, xgboost, imbalanced-learn, joblib
Development Environment	IDE	VS Code /pycharm
Data		
Data	Dataset	UCI Credit Card Approval Dataset / Kaggle Credit Card Dataset (CSV)